



Worksheet 2

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Branch: MCA(General)

Semester: III

Subject Name: Internet Of Things

UID: 25MCA20023

Section/Group: 25MCA-1-A

Date of Performance: 7/8/26

Subject Code: 25CAH-708

1. Aim/Overview of the practical:

Interface a led, pushbutton (**Pull Up**) and LCD with Arduino Uno. Use two led and two pushbuttons. Use pushbutton as toggle switch.

- Case 1: When press button 1 once then LED 1 will turn ON and print “**Button1 ON**” in Row 1 and “**LED 1 ON**” in Row 2.
- Case 2: When we press button 1 again then LED 1 will turn OFF and print “**Button1 OFF**” in Row 1 and “**LED 1 OFF**” in Row 2.
- Case 3: When we press button 2 once then LED 2 will turn ON and print “**Button2 ON**” in Row 1 and “**LED 2 ON**” in Row 2.
- Case 4: When we release button 2 again then LED 2 will turn OFF and print “**Button2 OFF**” in Row 1 and “**LED2 OFF**” in Row 2.

2. Apparatus

Hardware Requirements: Arduino Uno R3 Board

LCD

LED

Pushbutton

Software requirements: Microsoft Windows 10

TinkerCAD

3. Circuit Diagram(TinkerCad):

Case 1:

Coding:

```
#include <LiquidCrystal.h>
```

```
LiquidCrystal lcd(12, 11, 5, 4, 3, 2);
```

```
const int button1 = 7;
```

```
const int button2 = 8;
```

```
const int led1 = 9;
```

```
const int led2 = 10;
```

```
bool led1State = false;
```

```
int lastButton1 = HIGH;
int lastButton2 = HIGH;

void setup()
{
  pinMode(button1, INPUT_PULLUP);
  pinMode(button2, INPUT_PULLUP);

  pinMode(led1, OUTPUT);
  pinMode(led2, OUTPUT);

  lcd.begin(16, 2);
  lcd.clear();

  lcd.setCursor(0, 0);
  lcd.print("Ready");

  digitalWrite(led1, LOW);
  digitalWrite(led2, LOW);
}

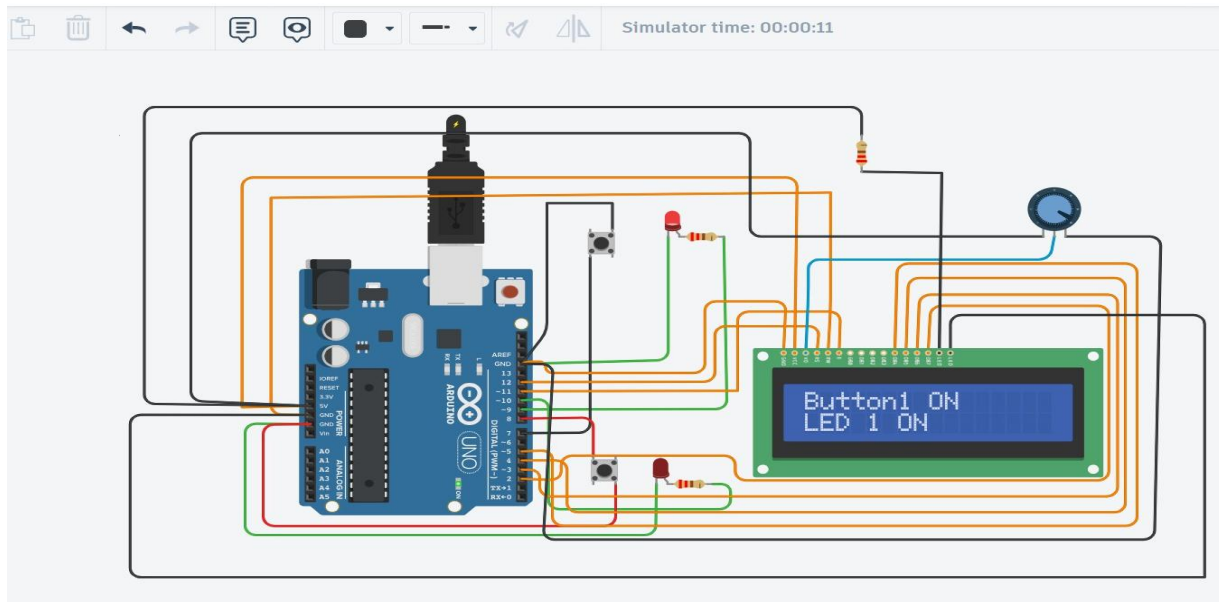
void loop()
{
  int b1 = digitalRead(button1);
  int b2 = digitalRead(button2);

  // BUTTON 1 PRESSED
  if (lastButton1 == HIGH && b1 == LOW)
  {
    led1State = !led1State;
    digitalWrite(led1, led1State);
    lcd.clear();

    if (led1State)
    {
      lcd.setCursor(0, 0);
      lcd.print("Button1 ON");

      lcd.setCursor(0, 1);
      lcd.print("LED 1 ON");
    }
    delay(200);
  }
  lastButton1 = b1;
}
```

Output:



Case 2:

Coding:

```
#include <LiquidCrystal.h>
LiquidCrystal lcd(12, 11, 5, 4, 3, 2);

const int button1 = 7;
const int button2 = 8;
const int led1 = 9;
const int led2 = 10;

bool led1State = false;

int lastButton1 = HIGH;
int lastButton2 = HIGH;

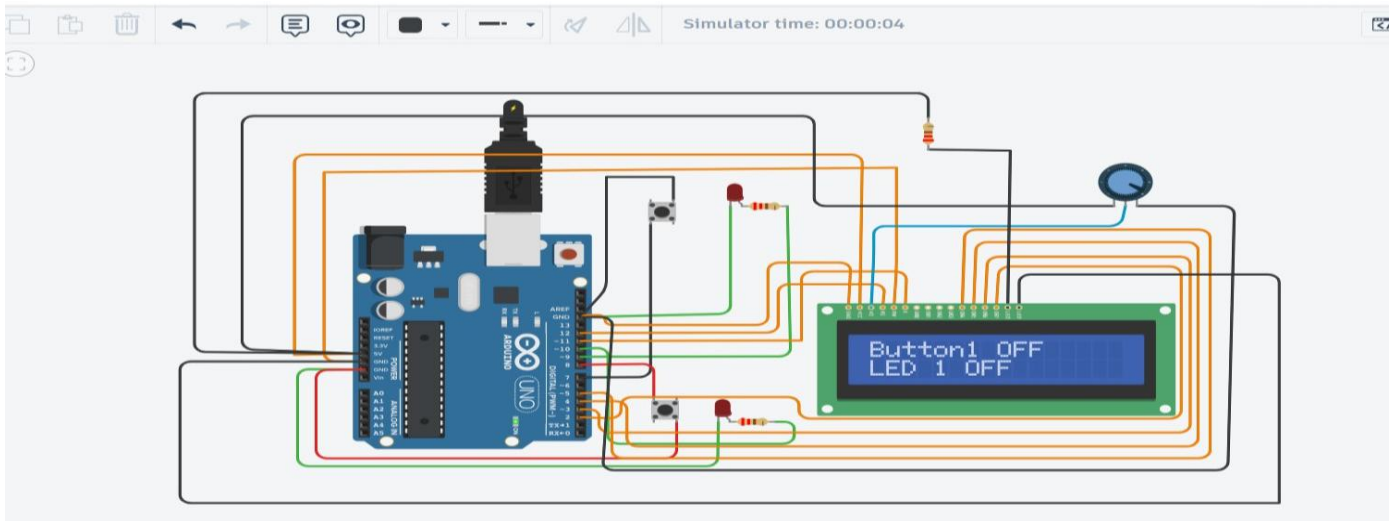
void setup()
{
  pinMode(button1, INPUT_PULLUP);
  pinMode(button2, INPUT_PULLUP);
  pinMode(led1, OUTPUT);
  pinMode(led2, OUTPUT);

  lcd.begin(16, 2);
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("Ready");
```

```
digitalWrite(led1, LOW);
digitalWrite(led2, LOW);
}
void loop()
{
  int b1 = digitalRead(button1);
  int b2 = digitalRead(button2);

  // BUTTON 1 PRESSED AGAIN
  if (lastButton1 == HIGH && b1 == LOW)
  {
    led1State = false;
    digitalWrite(led1, LOW);
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Button1 OFF");
    lcd.setCursor(0, 1);
    lcd.print("LED 1 OFF");
    delay(200);
  }
  lastButton1 = b1;
}
```

Output:



Case 3:

Coding:

```
#include <LiquidCrystal.h>
LiquidCrystal lcd(12, 11, 5, 4, 3, 2);
```

```
const int button1 = 7;
const int button2 = 8;
const int led1 = 9;
const int led2 = 10;

bool led1State = false;

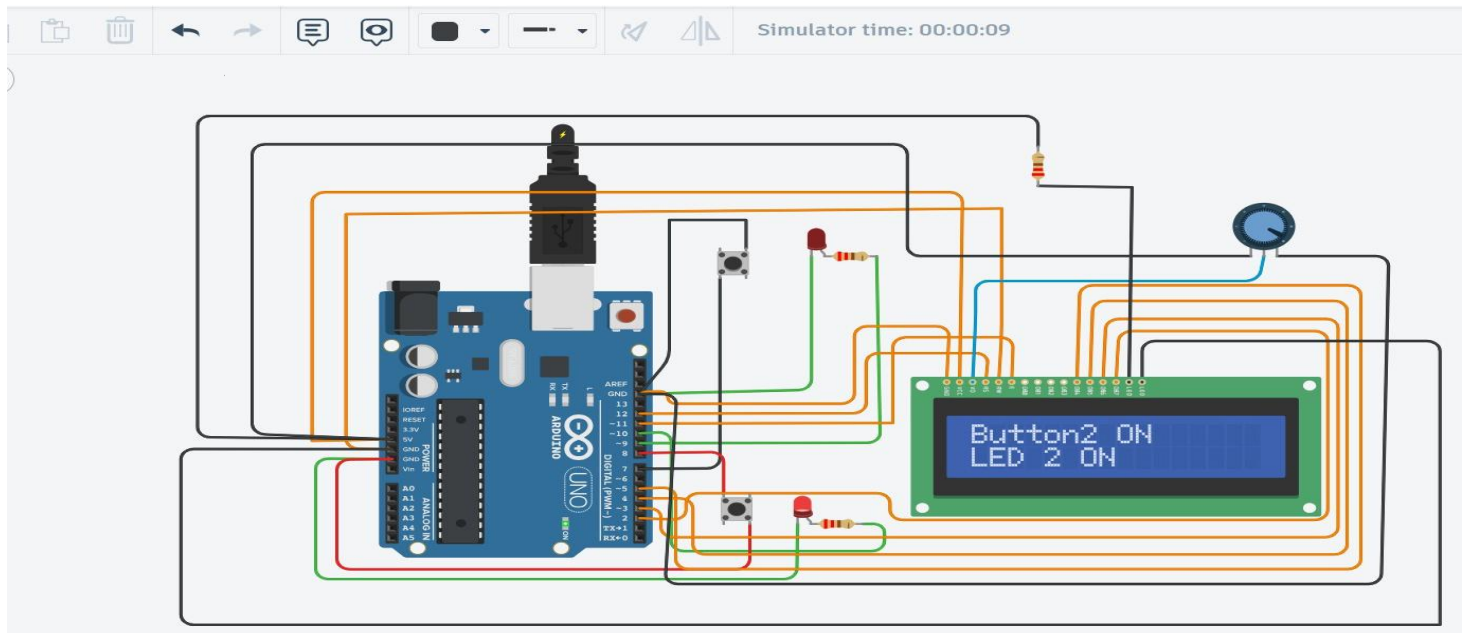
int lastButton1 = HIGH;
int lastButton2 = HIGH;
void setup()
{
  pinMode(button1, INPUT_PULLUP);
  pinMode (button2, INPUT_PULLUP);
  pinMode (led1, OUTPUT);
  pinMode (led2, OUTPUT);

  lcd.begin(16, 2);
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("Ready");
  digitalWrite(led1, LOW);
  digitalWrite(led2, LOW);
}

void loop()
{
  int b1 = digitalRead(button1);
  int b2 = digitalRead(button2);
  // BUTTON 2 PRESSED
  if (lastButton2 == HIGH && b2 == LOW)
  {
    digitalWrite(led2, HIGH);
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Button2 ON");
    lcd.setCursor(0, 1);
    lcd.print("LED 2 ON");
  }

  lastButton1 = b1;
  lastButton2 = b2;
  delay(50);
}
```

Output:



Case 4:

Coding:

```
#include <LiquidCrystal.h>
LiquidCrystal lcd(12, 11, 5, 4, 3, 2);
const int button1 = 7;
const int button2 = 8;
const int led1 = 9;
const int led2 = 10;

bool led1State = false;

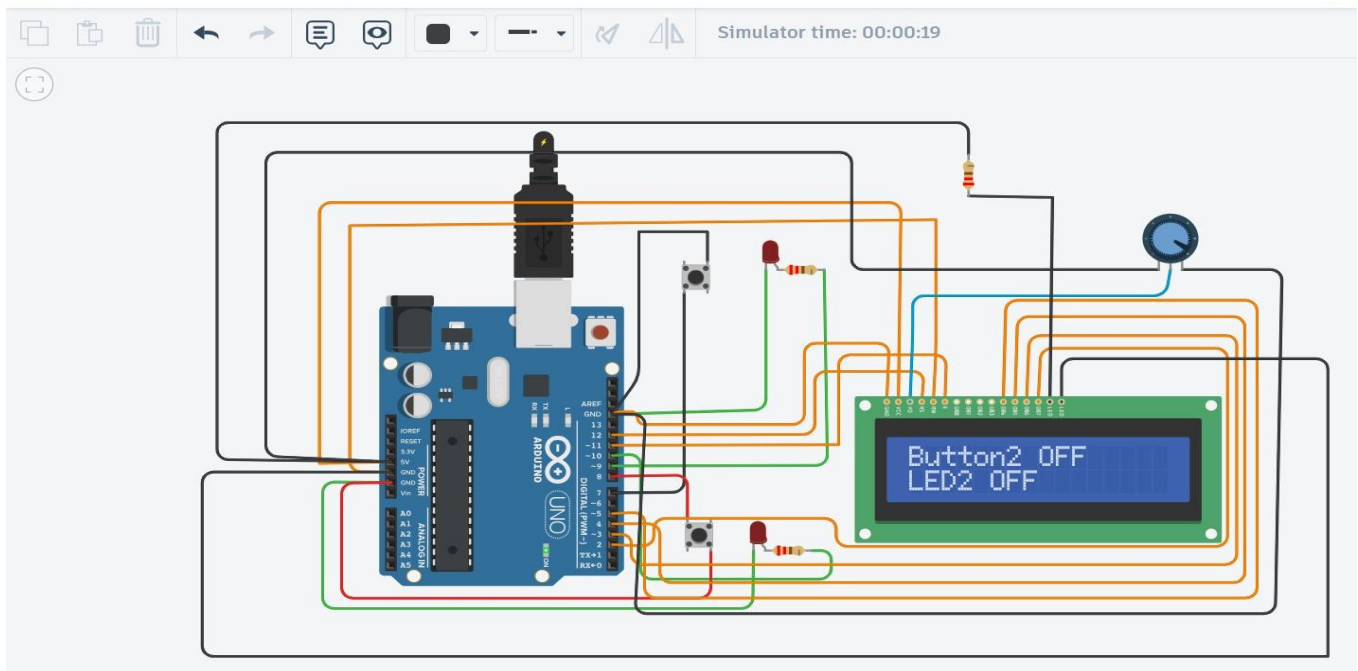
int lastButton1 = HIGH;
int lastButton2 = HIGH;
void setup()
{
    pinMode(button1, INPUT_PULLUP);
    pinMode(button2, INPUT_PULLUP);
    pinMode(led1, OUTPUT);
    pinMode(led2, OUTPUT);

    lcd.begin(16, 2);
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Ready");
```

```
digitalWrite(led1, LOW);
digitalWrite(led2, LOW);
}
void loop()
{
  int b1 = digitalRead(button1);
  int b2 = digitalRead(button2);

  // BUTTON 2 RELEASED
  if (lastButton2 == LOW && b2 == HIGH)
  {
    digitalWrite(led2, LOW);
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Button2 OFF");
    lcd.setCursor(0, 1);
    lcd.print("LED2 OFF");
  }
  lastButton1 = b1;
  lastButton2 = b2;
  delay(50);
}
```

Output:



5. Learning outcomes (What I have learnt):

1. Using a 16*2 LCD in circuit.
2. Circuit formation using Arduino uno R3, push button, LED, LCD and potentiometer.
3. Integrating a potentiometer in a circuit.
4. Concepts of pull up and pull down in push button.
5. Components of 16*2 LCD,